BMP Multiple User Handling with Role Types

Overview

This document explains how multiple users from the same BMP client are handled in Activepieces, including:

- **Shared Organizations**: Users with the same `clientId` automatically share organizations
- **Custom Platform Roles**: Users can be assigned different platform roles (ADMIN, MEMBER, OPERATOR, etc.)
- **SDK Authentication**: Using the `/auto-provision` endpoint for seamless user management

Table of Contents

1. Same ClientId Organization Sharing
2. Platform Roles
3. Custom Role Assignment
4. BMP Frontend Integration
5. API Examples
6. Use Cases

Same ClientId Organization Sharing

How It Works

When multiple users sign up or sign in with the **same `clientId` but different email addresses**, they will **automatically share the same organization**.

Flow Example

First User (User A):

1. Provides `clientId: "client-123"` and `clientName: "MyCompany"`
2. System stores `clientId` in User A's record
3. System creates organization "MYCOMPANY"
4. User A is assigned to this organization

Second User (User B):

1. Provides `clientId: "client-123"` (same as User A)
2. System stores `clientId` in User B's record
3. System checks: "Do any other users with `clientId: "client-123"` already have an organization?"
4. **YES** - User A already has organization "MYCOMPANY"
5. System assigns User B to the **same organization**
6. User B joins "MYCOMPANY" with User A

Implementation Details

```
// After storing clientId
const usersWithSameClientId = await userService.getByClientIdAndPlatform({
    clientId: params.clientId,
    platformId: params.platformId,
})

// Find existing organization from another user with same clientId
const existingOrgUser = usersWithSameClientId.find(
    u => u.id !== user.id && u.organizationId
)

if (existingOrgUser && existingOrgUser.organizationId) {
    // Assign new user to existing organization
    await userService.update({
        id: user.id,
        platformId: params.platformId,
        organizationId: existingOrgUser.organizationId,
    })
}
```

Important Notes

1. **Organization Name:** The organization name is determined by the **first user** who provides `clientName`. Subsequent users join that organization regardless of their `clientName` value.
2. **No `clientName` Provided:** If a user signs in without `clientName` but another user with the same `clientId` already has an organization, the user will be assigned to that existing organization.
3. **Platform Isolation:** Users with the same `clientId` only share organizations within the same `platformId`. Different platforms have separate organizations.

Platform Roles

Available Roles

Role	Description	Permissions
SUPER_ADMIN	System-wide administrator	Full system access
OWNER	Platform/tenant owner	Full platform access
ADMIN	Platform administrator	Can create projects, manage users
MEMBER	Regular platform member	Limited permissions, needs invites
OPERATOR	Platform operator	Automatic project access (except private)

Role Assignment Logic

Standard Signup (`/sign-up`)

- **With invitation**: Uses role from invitation (or defaults to MEMBER)
- **No invitation**: Defaults to **MEMBER**

Standard Login (`/sign-in`)

- **No role change** - User keeps existing platform role

SDK Authentication (`/auto-provision`)

- **Signup**: Assigns **ADMIN** role (or custom role via `roleName`)
- **Login**: User keeps existing role

Custom Role Assignment

The `roleName` Parameter

The `/auto-provision` endpoint supports custom platform roles via the **`roleName`** parameter.

Parameter Details:

- **Source**: `localStorage.getItem('ada.roleName')` from BMP frontend
- **Type**: String (case-insensitive)
- **Valid Values**: `SUPER_ADMIN` , `OWNER` , `ADMIN` , `MEMBER` , `OPERATOR`
- **Default**: `ADMIN` (if not provided or invalid)

Role Mapping Function

```
function mapRoleNameToPlatformRole(roleName?: string): PlatformRole {
    if (!roleName) {
        return PlatformRole.ADMIN // Default role
    }

    const normalizedRoleName = roleName.toUpperCase().trim()

    switch (normalizedRoleName) {
        case 'SUPER_ADMIN': return PlatformRole.SUPER_ADMIN
        case 'OWNER': return PlatformRole.OWNER
        case 'ADMIN': return PlatformRole.ADMIN
        case 'MEMBER': return PlatformRole.MEMBER
        case 'OPERATOR': return PlatformRole.OPERATOR
        default: return PlatformRole.ADMIN // Invalid role, default to ADMIN
    }
}
```

When Role is Applied

Signup Flow:

1. Creating auto-invitation → Uses `platformRole` from `roleName`
2. Creating user directly → Assigns `platformRole` from `roleName`
3. Defaults to ADMIN if `roleName` not provided

Login Flow:

- Role is **NOT changed** during login
- User keeps their existing role from signup

Important Notes

1. **Case Insensitive**: `"admin"`, `"ADMIN"`, `"Admin"` all map to `PlatformRole.ADMIN`
2. **Default Behavior**: Invalid or missing `roleName` defaults to `ADMIN` for backward compatibility
3. **Role Persistence**: Role is set during signup only; login does NOT change the user's role
4. **Security**: Role assignment happens server-side (client cannot bypass validation)

BMP Frontend Integration

JavaScript Example

```
// Get role and client info from localStorage
const roleName = localStorage.getItem('ada.roleName')      // e.g., "ADMIN", "MEMBER"
const clientName = localStorage.getItem('ada.clientName')  // e.g., "MyCompany"

// Prepare auto-provision request
const requestBody = {
  email: userEmail,
  password: userPassword,
  firstName: userFirstName,
  lastName: userLastName,
  platformId: '2Y6xAoWbvjiBgdRsBDcbP',  // Your platform ID
  clientId: 'client-123',                // Unique client identifier
  clientName: clientName,                // Organization name (optional)
  roleName: roleName                     // Custom role (optional)
}

// Call auto-provision endpoint
const response = await fetch('http://localhost:3000/api/v1/authentication/auto-provisio
  method: 'POST',
  headers: { 'Content-Type': 'application/json' },
  body: JSON.stringify(requestBody)
})

const data = await response.json()
// Store token and user info
localStorage.setItem('token', data.token)
```

Setting Role in BMP Frontend

```
// Set role before signup/login
localStorage.setItem('ada.roleName', 'MEMBER')    // For regular users
localStorage.setItem('ada.roleName', 'ADMIN')     // For administrators
localStorage.setItem('ada.roleName', 'OPERATOR')  // For operators
```

API Examples

Example 1: Team Admin with Custom Organization

```
POST /api/v1/authentication/auto-provision
{
  "email": "admin@company.com",
  "password": "secure123",
  "firstName": "Admin",
  "lastName": "User",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",
  "clientName": "MyCompany",
  "roleName": "ADMIN"
}

// Result: User created with ADMIN role in "MYCOMPANY" organization
```

Example 2: Team Member Joining Existing Organization

```
POST /api/v1/authentication/auto-provision
{
  "email": "member@company.com",
  "password": "secure456",
  "firstName": "Team",
  "lastName": "Member",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",      // Same clientId as admin
  "clientName": "MyCompany",
  "roleName": "MEMBER"
}

// Result: User created with MEMBER role, automatically joins "MYCOMPANY" organization
```

Example 3: Without Custom Role (Default ADMIN)

```
POST /api/v1/authentication/auto-provision
{
  "email": "user@example.com",
  "password": "password123",
  "firstName": "John",
  "lastName": "Doe",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "client-789",
  "clientName": "JohnOrg"
  // roleName not provided → defaults to ADMIN
}

// Result: User created with ADMIN role in "JOHNORG" organization
```

Example 4: Operator Role

```
POST /api/v1/authentication/auto-provision
{
  "email": "operator@company.com",
  "password": "secure789",
  "firstName": "Operator",
  "lastName": "User",
  "platformId": "2Y6xAoWbvjiBgdRsBDcbP",
  "clientId": "company-abc-123",      // Same clientId as others
  "clientName": "MyCompany",
  "roleName": "OPERATOR"
}

// Result: User created with OPERATOR role, joins "MYCOMPANY" organization
//         Has automatic access to non-private projects
```

Use Cases

Use Case 1: Company with Multiple Team Members

Scenario: A company has an admin, several members, and operators.

Setup:

```
// 1. Admin signs up first
{
  email: "admin@company.com",
  clientId: "company-123",
  clientName: "CompanyName",
  roleName: "ADMIN"
}

// 2. Team members join
{
  email: "member1@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "MEMBER"
}

{
  email: "member2@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "MEMBER"
}

// 3. Operator joins
{
  email: "ops@company.com",
  clientId: "company-123",  // Same clientId
  roleName: "OPERATOR"
}
```

Result:

- ✅ All users share the "COMPANYNAME" organization
- ✅ Admin can create and manage projects
- ✅ Members need project invites
- ✅ Operator has automatic access to projects

Use Case 2: Multi-Client SaaS Platform

Scenario: BMP frontend serves multiple client companies.

Client A:

```
{
  email: "user1@clienta.com",
  clientId: "client-a",
  clientName: "Client A Corp",
  roleName: "ADMIN"
}
```

Client B:

```
{
  email: "user1@clientb.com",
  clientId: "client-b",      // Different clientId
  clientName: "Client B Inc",
  roleName: "ADMIN"
}
```

Result:

- ✅ Client A users share one organization
- ✅ Client B users share a separate organization
- ✅ Complete isolation between clients
- ✅ Each can have their own role structure

Use Case 3: Gradual Team Growth

Scenario: Company starts with one user, adds more over time.

Timeline:

1. **Week 1**: First user signs up
   - Creates organization "MYCOMPANY"
   - Gets ADMIN role
2. **Week 2**: Second user joins
   - Same `clientId`
   - Automatically joins "MYCOMPANY"
   - Gets MEMBER role
3. **Week 3**: Third user joins
   - Same `clientId`
   - Automatically joins "MYCOMPANY"
   - Gets OPERATOR role

Result:

- ✅ Seamless team growth
- ✅ No manual organization assignment needed
- ✅ Each user has appropriate role
- ✅ All share same resources

Summary Table

Feature	Description
Endpoint	`/api/v1/authentication/auto-provision`
Organization Sharing	Same `clientId` → Same organization
Default Role	`ADMIN` (if `roleName` not provided)
Custom Roles	Via `roleName` parameter from localStorage
Role Options	SUPER_ADMIN, OWNER, ADMIN, MEMBER, OPERATOR
Platform Isolation	Different `platformId` → Separate organizations
Backward Compatible	✅ Yes - defaults to ADMIN

Database Schema

```
-- User table
user {
    id: string
    email: string (via identity)
    clientId: string (nullable)
    organizationId: string (nullable)
    platformId: string
    platformRole: enum (SUPER_ADMIN, OWNER, ADMIN, MEMBER, OPERATOR)
}

-- Organization table
organization {
    id: string
    name: string (e.g., "MYCOMPANY")
    platformId: string
}
```

Testing Checklist

- ✅ Same `clientId` with different emails → Users share organization
- ✅ Different `clientId` → Separate organizations
- ✅ `roleName: "ADMIN"` → User gets ADMIN role
- ✅ `roleName: "MEMBER"` → User gets MEMBER role
- ✅ `roleName: "OPERATOR"` → User gets OPERATOR role
- ✅ `roleName` not provided → Defaults to ADMIN
- ✅ Invalid `roleName` → Defaults to ADMIN
- ✅ Case-insensitive role names work correctly
- ✅ Login preserves existing role (no change)
- ✅ Platform isolation works (different platformId)

Benefits

1. **Automatic Team Management** - No manual organization assignment needed

2. **Flexible Role Structure** - Support for various permission levels
3. **Seamless Onboarding** - New users automatically join correct organization
4. **Client Isolation** - Different clients remain completely separate
5. **Backward Compatible** - Existing integrations continue to work
6. **Secure** - Server-side role validation and assignment

Date Created

February 16, 2026

Related Files

- Implementation: `/packages/server/api/src/app/authentication/authentication.controller.ts`
- User Service: `/packages/server/api/src/app/user/user-service.ts`
- User Entity: `/packages/server/api/src/app/user/user-entity.ts`